

Project Summary

Problem Statement

The rapid advancement of generative artificial intelligence has enabled modern deep learning models to synthesize highly realistic fake images, commonly referred to as deepfakes. While many existing deepfake detectors perform well on images generated by sources seen during training, they often fail to generalize to unseen generation models. Therefore, the primary goal of this project is to develop a **generalizable deepfake detection system** capable of identifying AI-generated images from multiple unseen generation sources.

The expected input is a single RGB image. The system predicts whether it is **real** or **fake/generated**. The output includes the predicted class label with confidence probabilities. The project focuses on reproducing and adapting the *Neighboring Pixel Relationships (NPR)* approach from the reference paper.

Approach

Our approach is based on the hypothesis that generative models introduce abnormal local pixel relationships due to repeated up-sampling. We implement the **NPR** transformation emphasizing local neighboring pixel differences.

Preprocessing: resize to 256×256 , center crop to 224×224 , normalize using ImageNet statistics, apply NPR with 2×2 *w1-reference*. The transformed image passes into a lightweight residual CNN (**NPRDetector**) for binary classification.

Pipeline:

Input $\rightarrow$ Preprocess $\rightarrow$ NPR $\rightarrow$ ResNet $\rightarrow$ Prediction

A deployment-ready inference pipeline was implemented using Gradio and Hugging Face Spaces.

Experiments and Results

We use the **ForenSynths/Wang-CVPR2020** dataset. Training used ProGAN four-class configuration: car, cat, chair, horse. Training splits had balanced real/fake samples.

Primary Phase 3 evaluation (8 sources): ProGAN, StyleGAN, StyleGAN2, BigGAN, CycleGAN, StarGAN, GauGAN, DeepFake

Pooled results:

Accuracy = 92.22%, AP = 97.31%, ROC-AUC = 97.58%

Source-wise comparison vs. reference paper (Table 1): **91.10%** vs. **92.45%** (gap of **1.35 pp**).

Supplementary experiments (2 additional GANs: BEGAN, SNGAN):

BEGAN: Accuracy = 91.47%, AP = 99.93%

SNGAN: Accuracy = 99.95%, AP = 100.00%

Combined mean: Accuracy = **95.71%**, AP = **99.96%**.

These experiments demonstrate that the implemented NPR-based detector generalizes effectively across multiple unseen GAN architectures and closely reproduces the reference paper's behavior.